

A two-replica proof of the Krenn–Gu conjecture

September 26, 2026

Abstract

In the graph model of photonic state preparation, the amplitude of a vertex colouring is a sum of products of complex edge weights over perfect matchings. The Krenn–Gu conjecture asks whether these weights can cancel every mixed colouring while retaining at least three monochromatic colourings on more than four vertices. We prove that they cannot. The main difficulty is that the existence of a mixed matching alone does not prevent its amplitude from cancelling. Our argument forces a stronger structure: after parallel edges with the same endpoint colours are aggregated, every edge is monochromatic and every active colour forms a perfect matching. A vertex colouring then has at most one contributing matching, so a known combinatorial obstruction finishes the proof. The structural argument uses two formal symbol families with identical pairing coefficients to encode the matching sums. Their symmetries impose polynomial identities that force each supported edge to contribute the entire amplitude of its colour. Complex weights, parallel edges, and distinct colours at the two ends of an edge are all allowed. A Lean formalization verifies the normalized equation-system statement at every even order at least six and every palette size at least three.

1 Introduction

Perfect matchings provide a bridge between graph theory and the preparation of entangled states with photon-pair sources [1]. Vertices represent output paths, edges represent pair contributions, and endpoint colours specify the modes. Selecting one photon at each output corresponds to selecting a perfect matching. Contributions that induce the same output modes add as complex amplitudes; they can therefore interfere destructively [2]. The resulting existence question was formulated by Krenn, Gu and Soltész [3, Question 1]; see also the matching examples in [4, Section 2].

For a Greenberger–Horne–Zeilinger (GHZ) target, the desired colourings assign the same colour to every vertex. All other colourings must have amplitude zero. Graph theory can force an unwanted matching to exist, but complex weights might cancel its contribution against other matchings. The proof must therefore explain why such cancellation cannot satisfy all the target equations simultaneously. We do this by deriving a structural restriction on the weighted graph itself.

1.1 Matching amplitudes and the conjecture

Let $d \geq 1$ and write $[d] = \{1, \dots, d\}$ for the target palette. Let $G = (V, E)$ be a finite loopless multigraph. Each edge $e = uv$ has weight $w(e) \in \mathbb{C}$ and two endpoint colours $\chi_u(e), \chi_v(e)$ in $[d]$. A *perfect matching* is a set of edges in which every vertex is incident to exactly one edge. We write $\mathcal{M}(G)$ for the set of perfect matchings of G . For $M \in \mathcal{M}(G)$, let $e_M(v)$ be its unique edge at v . The matching induces the vertex colouring $v \mapsto \chi_v(e_M(v))$. For a vertex colouring $\kappa : V \rightarrow [d]$,

define

$$W(\kappa) = \sum_{\substack{M \in \mathcal{M}(G) \\ \chi_v(e_M(v)) = \kappa(v) \text{ for every } v}} \prod_{e \in M} w(e).$$

An empty sum is zero. We write i^V for the colouring that is constantly i . A colouring that is not constant is called mixed. For a matrix B and a vertex set S , $B[S]$ denotes its principal submatrix on S .

In the physical interpretation, these coefficients give the unnormalized postselected state

$$|\psi_G\rangle = \sum_{\kappa: V \rightarrow [d]} W(\kappa) \bigotimes_{v \in V} |\kappa(v)\rangle_v.$$

Here $|h\rangle_v$ denotes basis mode h in output path v . Postselection means conditioning on one photon in each path. The weights and their matching products are amplitudes, not probabilities. The theorem concerns the states represented by this matching model.

Definition 1.1 (Generalized GHZ condition). The weighted half-edge-coloured graph satisfies the generalized GHZ condition on $[d]$ if $W(i^V) = \tau_i \neq 0$ for every $i \in [d]$, and $W(\kappa) = 0$ for every mixed vertex colouring $\kappa : V \rightarrow [d]$.

The usual normalized condition requires $\tau_i = 1$. Our definitions follow Chandran, Gajjala and Illickan [5, Sections 1.1 and 1.4]; the generalized condition permits unequal nonzero amplitudes. We call a colour *active* when its monochromatic amplitude is nonzero. All colours in the target palette of Definition 1.1 are therefore active.

Theorem 1.2 (Krenn–Gu conjecture). *If $|V| > 4$ is even, there is no complex weighted half-edge-coloured multigraph satisfying the generalized GHZ condition on a palette of size $d \geq 3$.*

The theorem concerns this matching model with arbitrary complex weights. There is no restriction on edge density or on the number of parallel edges. It suffices to retain three active colours: removing edges with other endpoint colours leaves every amplitude on the retained palette unchanged.

1.2 Cancellation and the four-vertex boundary

Consider the left-hand graph in Figure 1. Both of its perfect matchings induce the mixed colouring (a, a, b, b) . Their weights are $1 \cdot 1$ and $1 \cdot (-1)$, so

$$W(a, a, b, b) = 1 \cdot 1 + 1 \cdot (-1) = 0.$$

Thus a zero amplitude does not imply the absence of a matching. This small graph illustrates cancellation only; it does not satisfy the GHZ condition. The proof has to use the nonzero monochromatic amplitudes together with the vanishing of *every* mixed amplitude.

The restriction to more than four vertices is necessary. In Figure 1(b), the three perfect matchings of K_4 are

$$\{12, 34\}, \quad \{13, 24\}, \quad \{14, 23\}.$$

Colour these matchings a, b, c , respectively, and give every edge weight one. The only induced colourings are a^V, b^V, c^V , each with amplitude one. Our structural argument will recover precisely the feature visible in this example: each colour forms a perfect matching. The final graph obstruction then distinguishes four vertices from larger orders.

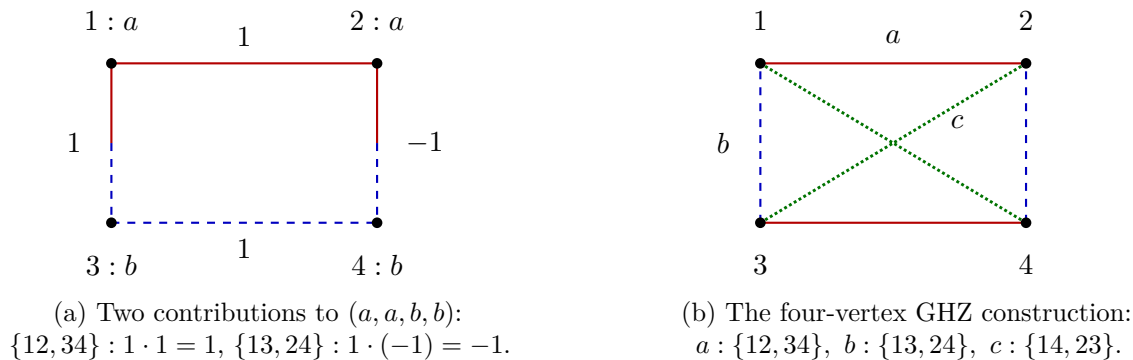


Figure 1: Cancellation and the four-vertex exception. Panel (a) illustrates cancellation only and does not satisfy the GHZ condition. Labels beside edges are weights and labels at vertices give the inherited colours; the two vertical edges have different colours at their ends. In (b), all weights are one. Horizontal, vertical, and diagonal edges have colours a, b, c , respectively. Solid, dashed, and dotted strokes also distinguish these colours. The diagonal crossing is not a vertex.

1.3 Related work and contribution

The graph correspondence originates with Krenn, Gu and Zeilinger [1]; its weighted interference formulation is developed further in [2]. The weighted, endpoint-coloured existence question is stated in [3, Question 1]. Complex amplitudes distinguish this question from the purely combinatorial problem of forbidding mixed perfect matchings.

Bogdanov proved the combinatorial obstruction in 2017 [6]; [5, Theorem 7] records the form we use. Chandran and Gajjala classify the simple graphs admitting edge colourings in which every perfect matching is monochromatic, and derive weighted consequences [7]. Their separate sparsification theorem excludes dimension $d \geq n/\sqrt{2}$ on simple graphs with $n > 4$, permitting complex weights and bichromatic edges [8, Theorem 8]. In contrast, Cervera-Lierta, Krenn and Aspuru-Guzik’s SAT exclusions at $(n, d) = (6, 3)$ and $(8, 4)$ assume monochromatic edges. Their general Boolean conditions are necessary support constraints; satisfiability alone does not construct the complex amplitudes [4, Sections 3.1–3.2].

In the general weighted multigraph model, Chandran, Gajjala and Illickan prove the conjecture when the underlying simple graph has vertex connectivity at most two or maximum degree at most three. Their reduction also forces a minimum-vertex counterexample to be four-connected [5, Theorems 9–11]. Their cubic-case proof already obtains a unique matching for each contributing colouring and then applies Bogdanov’s obstruction [5, Section 2.3, Corollary 17]. We use that same final strategy. The contribution here is the algebraic derivation of monochromatic aggregate edges and degree one in every active colour without a sparsity assumption. Further work by Chandran and Gajjala on connectivity bounds is accepted for FSTTCS 2026 [9]; only the conference listing was available in the sources consulted, so we do not compare theorem statements or methods here.

Recent formal work addresses further cases. A public Lean certificate artifact exports the complex six-vertex, three-colour statement [10]. Krenn’s problem page reports an all-even-order obstruction for $d \geq n$ in 2026 [11]. The AlphaProof Nexus preprint discusses this research and explicitly lists $N = d \in \{4, 6, 10\}$, while citing a follow-up tensor-algebraic manuscript as in preparation [12, Section 4 and reference 38]. We distinguish the author-announced bound from the displayed instances in that preprint; the unavailable follow-up manuscript is not an input to our proof or a basis for a comparison of methods.

The standard algebraic ingredients are the pairing formula, polarization, and scalar orthogonal invariant theory [13, 14]. We define and explain their use below. The matching equations supply the additional response identities and matrix divisibility that are needed for the supported-edge identity.

1.4 Proof overview: forcing one edge per colour

We first remove an ambiguity caused by parallel edges. For fixed vertices u, v and fixed endpoint colours i, j , replace all corresponding parallel edges by one edge whose weight is their sum. Delete that edge if the sum is zero. A perfect matching uses at most one edge between u and v , so distributivity shows that this operation preserves every $W(\kappa)$. All support and uniqueness statements below refer to this *aggregate graph*. This reduction is also stated in [5, page 41:5].

The argument has two structural steps and a short combinatorial conclusion. We describe the conclusion first far enough to make the role of each structural step explicit.

Step 1: every remaining edge has equal endpoint colours. Lemma 4.2 proves that every aggregate weight with distinct endpoint colours is zero. After this reduction, fix an active colour and write B_{pq} for its weight on the edge pq . A supported edge means one with $B_{pq} \neq 0$. Let $\beta \neq 0$ be the amplitude of the monochromatic colouring in this colour.

Step 2: each supported edge contributes the entire amplitude. For an edge pq , define its matching subtotal

$$C_{pq} = \sum_{\substack{M \text{ a perfect matching of this colour} \\ pq \in M}} \prod_{e \in M} w(e).$$

The central identity, proved in Theorem 5.1, is

$$\boxed{C_{pq} = \beta \quad \text{whenever } B_{pq} \neq 0.}$$

Its consequence is immediate. Every perfect matching uses exactly one edge at a fixed vertex p . If $\deg_B(p)$ counts the supported edges of this colour at p , then

$$\beta = \sum_{q: B_{pq} \neq 0} C_{pq} = \deg_B(p) \beta.$$

Since $\beta \neq 0$, we obtain $\deg_B(p) = 1$. This applies to every vertex and every active colour: each colour graph is a perfect matching. In particular, the conclusion uses complex amplitudes directly and requires no positivity argument.

Step 3: a mixed matching can no longer cancel. Given a vertex colouring κ , its colour at each vertex specifies the only possible incident edge in a contributing matching. Consequently there is at most one such matching. If it exists, its weight is a product of nonzero edge weights and hence is nonzero. But three monochromatic perfect matchings of distinct colours on more than four vertices force a mixed perfect matching [5, Theorem 7]. Its nonzero amplitude contradicts the GHZ condition. We include the graph argument in Section 7.

Where the algebra enters. The difficult work is to establish Steps 1 and 2. A matching subtotal can be written as

$$C_{pq} = B_{pq} \text{haf}(B[V \setminus \{p, q\}]),$$

where the hafnian is the sum of products over perfect matchings on the remaining vertices. Thus Step 2 is an identity relating a full matching sum to a sum with two vertices removed. The remaining graph is not assumed to satisfy any GHZ condition; the identity must follow from the equations on the original graph.

We obtain such identities by taking two identical formal copies of the matching polynomial. Mixing the copies preserves their pairing data, while certain alternating expressions change sign. This yields vanishing identities that are not visible in a single matching sum. A second use of the symmetry restricts an auxiliary matrix to a small family of polynomial forms. The original matching equations then impose differential equations on their coefficients. A polynomial of positive degree cannot satisfy $p' + \delta p = k$ when $\delta \neq 0$ and k is constant: its highest-degree term would have nothing to cancel. This finite-degree argument gives the supported-edge identity.

Section 2 develops the pairing language from a four-factor example. Sections 3 and 4 establish Step 1; Section 5 proves the identity in Step 2. The final two steps above are completed in Sections 6 and 7. Section 8 states the precise formalization scope.

2 Algebraic formulation and formal pairings

2.1 Encoding a matching sum

Our first algebraic operation enforces the matching rule: a vertex may be used only once. Associate a variable $x_{v,i}$ with colour i at vertex v , and set every product $x_{v,i}x_{v,j}$ to zero. Thus a product of edges sharing a vertex vanishes, regardless of their colours. Products of vertex-disjoint edges survive.

Write $n = |V|$ and fix an ordering of V . Let $A_{uv}(i, j)$ be the sum of the weights of edges whose endpoint colours are i at u and j at v , as in the aggregation in Section 1.4. When we reverse the vertex order, we use $A_{vu}(j, i) = A_{uv}(i, j)$. The algebra and its edge polynomial are

$$\begin{aligned} A_V &= \mathbb{C}[x_{v,i} : v \in V, i \in \{a, b, c\}] / (x_{v,i}x_{v,j} : v \in V, i, j \in \{a, b, c\}), \\ A &= \sum_{u < v} \sum_{i, j \in \{a, b, c\}} A_{uv}(i, j) x_{u,i} x_{v,j}. \end{aligned}$$

Every surviving term in its divided top power chooses each vertex exactly once. The division by $(n/2)!$ removes the ordering of the chosen edges. Thus the coefficient of $\prod_v x_{v,\kappa(v)}$ is exactly $W(\kappa)$. We use $A^{[j]} = A^j/j!$ for these divided powers. For one colour on four vertices, the coefficient is the familiar three-term matching sum

$$\text{haf}(B) = B_{12}B_{34} + B_{13}B_{24} + B_{14}B_{23}.$$

In general, $\text{haf}(B[S])$ denotes the corresponding sum on the even vertex set S , with value one when S is empty.

We may therefore express the ternary instance of Definition 1.1 as follows. Write a_v, b_v, c_v for the three coordinates at v , and $h^S = \prod_{v \in S} h_v$.

Theorem 2.1 (Algebraic form of Theorem 1.2). *Let V have even cardinality $n > 4$. In the algebra $\mathcal{A}_V$, let A be any quadratic involving only distinct vertices, with arbitrary complex coefficients. Then*

$$A^{[n/2]} = \tau_a a^V + \tau_b b^V + \tau_c c^V, \quad \tau_a \tau_b \tau_c \neq 0 \quad (1)$$

is impossible.

The coefficient interpretation just proved identifies this statement with the graph theorem after palette restriction. The algebra keeps the original complex weights, including all aggregate bichromatic weights.

2.2 Rows, colour words, and coefficients

We use the same construction on any subset $S \subseteq V$: $\mathcal{A}_S$ has variables $x_{v,h}$ for $v \in S$ and $h \in \{a, b, c\}$, with the same rule that two variables at one vertex have product zero. Its degree-one elements are called *rows*. Thus

$$L = \sum_{v \in S} \sum_{h \in \{a, b, c\}} L_v(h) x_{v,h}, \quad L_v(h) \in \mathbb{C}.$$

A row is a list of coefficients, one for each vertex and colour, written as a linear polynomial. The term comes from the edges incident on one fixed vertex; the precise incident row is defined in Section 3.1.

A *colour word* on S is simply a colouring $\kappa : S \rightarrow \{a, b, c\}$. It labels the monomial $x^\kappa = \prod_{v \in S} x_{v, \kappa(v)}$. These monomials form a basis of the degree- $|S|$ part of $\mathcal{A}_S$, which we denote by $\mathcal{T}_S$. An element $F \in \mathcal{T}_S$ is a finite array of colouring coefficients, also called a *tensor* in the proof. We write $[\kappa]F$, or equivalently $F(\kappa)$, for the coefficient of x^κ in F . This is coefficient extraction, not evaluation of the polynomial at a point. In particular, $[h^S]F$ selects the constant colouring of colour h .

A *binary projection* retains only two chosen colours at every vertex: set the variables of the third colour to zero. A *binary word* uses at most those two colours. These operations will let us test selected coefficients of the original three-colour equations. We denote the projection onto colours h, k by π_{hk} .

2.3 Normalizing nonzero target amplitudes

Lemma 2.2 (Normalization of pure amplitudes). *A fixed half-edge-coloured graph with at least one vertex admits weights satisfying the generalized GHZ condition if and only if it admits weights satisfying the normalized condition with every pure amplitude equal to one. The rescaling between these assignments preserves which edge weights are nonzero.*

Proof. The written argument permits arbitrary nonzero τ_i ; the formal interface uses unit pure amplitudes. To relate them, fix a vertex r . Multiply each edge incident on r , with colour i at that endpoint, by τ_i^{-1} . Every perfect matching contains exactly one such edge, so the new amplitude is

$$W'(\kappa) = \tau_{\kappa(r)}^{-1} W(\kappa).$$

Every pure amplitude becomes one and every mixed amplitude remains zero. The converse follows directly from the generalized definition. $\square$

This is an alternative realization of the normalization equivalence in [5, Lemma 14], obtained by rescaling at just one vertex. It changes edge weights and preserves existence. It is distinct from dividing a quantum state by its norm and need not be a unitary operation.

2.4 Formal pairings: a four-factor example

A second algebraic description lets us exploit symmetry. Start with an index set Λ , formal commuting symbols Z_λ for $\lambda \in \Lambda$, and a symmetric table of complex numbers $C_{\lambda\mu} = C_{\mu\lambda}$. Declare the *pairing* $\langle Z_\lambda, Z_\mu \rangle_C = C_{\lambda\mu}$, and extend it bilinearly to linear combinations of symbols. Bilinear means linear in both arguments, without complex conjugation.

On the ordinary polynomial ring $\mathbb{C}[Z_\lambda : \lambda \in \Lambda]$, define a linear functional $\mathcal{W}_C$, called the *formal Wick functional*, by summing over pairings of the labelled factor occurrences:

$$\mathcal{W}_C(Z_{\lambda_1} \cdots Z_{\lambda_{2r}}) = \sum_{M \in \mathcal{M}(\{1, \dots, 2r\})} \prod_{\{i, j\} \in M} C_{\lambda_i \lambda_j}.$$

Here $\mathcal{M}(\{1, \dots, 2r\})$ is the set of partitions of the $2r$ positions into unordered pairs, equivalently the perfect matchings on those positions. For four factors this definition says

$$\mathcal{W}_C(Z_1 Z_2 Z_3 Z_4) = C_{12} C_{34} + C_{13} C_{24} + C_{14} C_{23}.$$

An odd number of factors has value zero, and $\mathcal{W}_C(1) = 1$. Repeated symbols still occupy distinct positions in the pairing sum.

The table entries are traditionally called *covariances*, and values of $\mathcal{W}_C$ are called *moments*. Here both terms refer only to the displayed finite sums. No probability distribution, expectation, positivity, or invertibility is assumed. We can also use polynomial parameters as entries and apply the same definition coefficientwise. We use the pairing polynomial appearing in Isserlis's formula [13, Section 2, equations (5)–(6)] as an algebraic definition. The extension to arbitrary complex pairing tables is part of this definition; it does not require a probabilistic realization of the table.

For a quadratic matching polynomial R on a vertex set S , take $\Lambda = S \times \{a, b, c\}$. The entry $C_{(v,h),(w,k)}$ is the coefficient of $x_{v,h} x_{w,k}$ in R when $v \neq w$, and is zero when $v = w$. For even $|S| = 2r$,

$$[\kappa] R^{[r]} = \mathcal{W}_C \left(\prod_{v \in S} Z_{v, \kappa(v)} \right).$$

Both sides enumerate the same matchings. This identity is the bridge between the two algebraic descriptions.

The two kinds of variables have different rules. Lower-case $x_{v,h}$ belong to $\mathcal{A}_S$ and obey the one-use-per-vertex rule. Upper-case $Z_{v,h}$ are ordinary formal symbols and obey no such relation. For example, if $\langle Z_u, Z_v \rangle_C = c$ and the two self-pairings are zero, then $\mathcal{W}_C(Z_u^2 Z_v^2) = 2c^2$, although $x_u^2 x_v^2 = 0$ in the site algebra. We compare the descriptions through coefficient identities such as the one above; we do not apply the Wick functional to the quotient algebra $\mathcal{A}_S$.

2.5 Why take two copies?

Here a *copy*, also called a *replica*, is a second set of formal symbols carrying the same pairing table. Introduce two disjoint families X_λ and Y_λ , with the same index set Λ as above, in an ordinary

polynomial ring. Define their joint pairing by

$$\begin{array}{c|cc} \langle \cdot, \cdot \rangle_{\tilde{C}} & X_\mu & Y_\mu \\ \hline X_\lambda & C_{\lambda\mu} & 0 \\ Y_\lambda & 0 & C_{\lambda\mu} \end{array} \quad \tilde{C} = \begin{pmatrix} C & 0 \\ 0 & C \end{pmatrix}.$$

Thus two symbols in the same family have their original pairing coefficient; symbols from different families have pairing coefficient zero. The joint Wick functional $\mathcal{W}_{\tilde{C}}$ is defined by the same sum-over-pairings rule. These copies are bookkeeping devices for multiplying and comparing matching sums. They are not additional photons, physical experiments, or copies of a quantum state. In particular, cross-family pairings contribute zero, so $\mathcal{W}_{\tilde{C}}(F(X)G(Y)) = \mathcal{W}_C(F)\mathcal{W}_C(G)$ for polynomials F, G .

A *copy transformation* mixes the two symbols at each fixed label, using the same matrix at every label. For example,

$$X'_\lambda = \frac{X_\lambda + Y_\lambda}{\sqrt{2}}, \quad Y'_\lambda = \frac{X_\lambda - Y_\lambda}{\sqrt{2}}$$

preserves the joint pairing table. Indeed, bilinearity gives

$$\langle X'_\lambda, X'_\mu \rangle_{\tilde{C}} = \frac{1}{2}(C_{\lambda\mu} + C_{\lambda\mu}) = C_{\lambda\mu}, \quad \langle X'_\lambda, Y'_\mu \rangle_{\tilde{C}} = \frac{1}{2}(C_{\lambda\mu} - C_{\lambda\mu}) = 0,$$

and the Y', Y' pairing is also $C_{\lambda\mu}$. Because every Wick sum is built from these entries, every Wick sum is preserved.

In general the allowed matrices form $O(2, \mathbb{C}) = \{O \in \mathbb{C}^{2 \times 2} : O^T O = I_2\}$, where T means transpose and I_2 is the two-dimensional identity matrix. This uses no complex conjugation; it is an orthogonal symmetry of a complex bilinear pairing, rather than a unitary transformation of states. We write $SO(2, \mathbb{C})$ for those matrices with determinant one, and suppress $\mathbb{C}$ in the group notation below. The index $\lambda = (v, h)$, the graph, and its edge weights stay fixed. Figure 2 summarizes the construction.

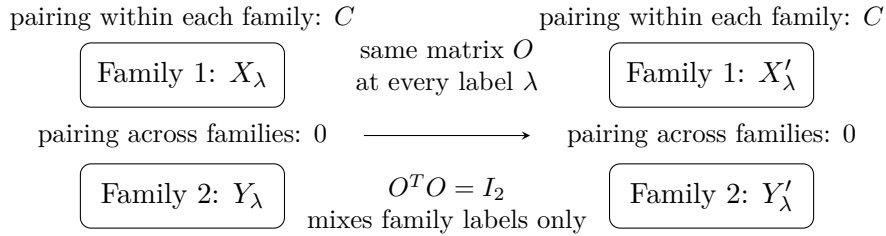


Figure 2: Two formal copies of one pairing table. Each box represents a whole family of symbols indexed by the same vertex-colour labels λ . The arrow denotes a change of formal variables, not a graph edge or a physical operation. The joint table $\tilde{C}$, and hence every Wick sum computed from it, is unchanged.

There is also an expression that detects the determinant of this change of copy coordinates. For a vertex v , choose distinct colours α_v, β_v and use $X_{v,h} = X_{(v,h)}$, $Y_{v,h} = Y_{(v,h)}$. Form

$$\Delta_v = X_{v,\alpha_v} Y_{v,\beta_v} - X_{v,\beta_v} Y_{v,\alpha_v}.$$

This is the determinant of the matrix with rows $(X_{v,\alpha_v}, X_{v,\beta_v})$ and $(Y_{v,\alpha_v}, Y_{v,\beta_v})$. Under a copy transformation O it becomes $(\det O)\Delta_v$. A reflection, meaning an orthogonal matrix of determinant -1 , changes the sign of a product of Δ_v 's on an odd set of vertices. If the remaining factors in a

moment are fixed by that reflection, invariance of the moment forces it to vanish. Section 3 chooses those factors and the reflection to obtain identities for the original incident rows.

On an even set of vertices the product instead stays unchanged. This gives the invariant pairing used in Section 5. Its polynomial dependence on auxiliary parameters is essential: degree comparisons turn the symmetry constraints into exact identities. All parameter variations take place in these auxiliary expressions, with the original edge weights held fixed. We never assume that deleting vertices preserves the GHZ condition.

2.6 Shifted pairing sums and an alternating pairing

For a row L on S , replacing $Z_{v,h}$ by $Z_{v,h} + L_v(h)$ adds a scalar to each formal symbol. We call L the *shift row*; the traditional word *mean* will refer only to these auxiliary scalars. For a fixed quadratic R , define

$$\mathcal{F}_{S,R}(L) = \sum_{j=0}^{\lfloor |S|/2 \rfloor} \frac{L^{|S|-2j}}{(|S|-2j)!} R^{[j]} \in \mathcal{T}_S.$$

For any colour word κ , expansion by the factors that supply a scalar shift gives

$$[\kappa]\mathcal{F}_{S,R}(L) = \mathcal{W}_C \left(\prod_{v \in S} (Z_{v,\kappa(v)} + L_v(\kappa(v))) \right).$$

Choose the shifted vertices, then pair the remaining vertices. This counts each contribution once on both sides; the factorials on the left remove the ordering of the selected row factors and edges. The odd and even cases of this polynomial will be denoted by Ψ and E , respectively, in the following sections.

Fix an ordered pair of colours (a, b) . Define a local bilinear pairing by

$$\epsilon(a, b) = 1, \quad \epsilon(b, a) = -1, \quad \epsilon(a, a) = \epsilon(b, b) = 0.$$

On binary words, define

$$K_S(x^\kappa, x^\lambda) = \prod_{v \in S} \epsilon(\kappa(v), \lambda(v)),$$

and extend bilinearly to coefficient arrays in $\mathcal{T}_S$, first projecting onto colours a, b . Thus only complementary words contribute. If $\bar{\kappa}$ exchanges a and b in κ , the explicit formula is

$$K_S(F, G) = \sum_{\kappa: S \rightarrow \{a, b\}} (-1)^{|\kappa^{-1}(b)|} [\kappa]F[\bar{\kappa}]G.$$

In particular, $K_S(a^S, b^S) = 1$ and $K_S(G, F) = (-1)^{|S|} K_S(F, G)$. The pairing is symmetric for even $|S|$ and alternating for odd $|S|$. It is bilinear, not Hermitian. We write K when the vertex set and ordered colour pair have been specified.

Expanding $\prod_{v \in S} \Delta_v$ produces exactly these signs. Together with the shifted-Wick identity, this explains why a simultaneous copy transformation acts on both shift rows in $K_S(\mathcal{F}_{S,R}(L), \mathcal{F}_{S,R}(M))$. For even $|S|$ this polynomial is invariant under $O(2)$; for odd $|S|$ it is invariant under $SO(2)$.

All derivatives below are derivatives of finite polynomials in shift parameters. For a polynomial map E on rows and fixed directions J, J' , our convention is

$$E_J(L) = \left. \frac{d}{dt} E(L + tJ) \right|_{t=0}, \quad E_{JJ'}(L) = \left. \frac{\partial^2}{\partial s \partial t} E(L + sJ + tJ') \right|_{s=t=0}.$$

The underlying quadratic and all graph weights are held fixed.

3 Vanishing identities after deleting one vertex

3.1 Incident rows and their responses

A *root* is a vertex singled out for a matching expansion. Delete a root p from the original graph and put $\Omega = V \setminus \{p\}$, so $|\Omega| = 2m + 1$. Decompose the edge polynomial as

$$A = R + \sum_{h \in \{a,b,c\}} x_{p,h} L_h^{(p)}, \quad L_h^{(p)} = \sum_{v \neq p} \sum_k A_{pv}(h,k) x_{v,k}.$$

The quadratic R contains the edges avoiding p ; the row $L_h^{(p)}$ lists the ways to leave p with colour h at that endpoint. The coefficient of $x_{p,h}$ in the target equation is

$$L_h^{(p)} R^{[m]} = \tau_h h^\Omega.$$

We call $LR^{[m]}$ the *response* of the row L : it selects one incident connection and matches the remaining vertices with edges of R . This use of “response” refers to a polynomial construction, not to a physical susceptibility.

For odd p , the higher response $L^p R^{[(2m+1-p)/2]}$ inserts p row factors and matches the other vertices through R . Here and in the next lemma p is an insertion count, not the name of a graph vertex. These higher expressions are auxiliary identities; they do not represent repeated physical use of a deleted root. The largest insertion count, $p = 2m + 1$, is called *terminal*: its power of R is zero.

The following lemma says that every higher response vanishes on every binary word. Its proof has three ingredients: a reflection relates its coefficients; rotations constrain the remaining common factor; finite degree makes that factor constant.

3.2 The vanishing lemma

Lemma 3.1 (Binary higher-response vanishing). *Let Ω have $2m + 1 \geq 3$ vertices, let R be an arbitrary quadratic in $\mathcal{A}_\Omega$, and let $D \subseteq (\mathcal{A}_\Omega)_1$ be a complex vector space of rows L such that*

$$\Phi(L) = LR^{[m]} = \sum_h f_h(L) h^\Omega. \tag{2}$$

Here the linear function $f_h : D \rightarrow \mathbb{C}$ extracts the pure colour coefficient of $\Phi(L)$. Assume all three f_h are nonzero functions and at least two are linearly independent. For odd $p \leq 2m + 1$, define

$$H_p(L) = L^p R^{[(2m+1-p)/2]}.$$

For every odd $3 \leq p \leq 2m + 1$, the tensor $H_p(L)$ has zero coefficient on every word using at most two colours, including the terminal degree $p = 2m + 1$. The same vanishing holds after polarization for products of arbitrary rows from the same D .

We will use ordinary polynomials in the coordinates of $L \in D$. If $L = \sum_j t_j L_j$ in a basis of D , this ring is $\mathbb{C}[D] = \mathbb{C}[t_1, \dots, t_{\dim D}]$. A nonzero coefficient function can be cancelled in this ring even if it vanishes at some individual rows. Such cancellation is never performed in the site algebra, which has zero divisors.

Proof. Pairing representation and reflection. Give formal variables $X_{i,h}$ their R covariances, zero covariances within a vertex, and give an auxiliary symbol ζ covariance $L_i(h)$ with $X_{i,h}$ and zero covariance with itself. A Wick moment with p occurrences of ζ and one specified coordinate at each vertex is the corresponding coefficient of $H_p(L)$. The $p!$ assignments of auxiliary occurrences are precisely those of the raw power L^p ; the divided power of R counts each remaining matching once.

Duplicate this entire symbol family, including ζ , to obtain $(X, \zeta_1), (Y, \zeta_2)$ with zero cross-family pairings as in Section 2.5. The same copy matrix acts on every symbol, including the two auxiliary symbols. Choose distinct local colours α_i, β_i at each vertex and set

$$\Delta_i = X_{i,\alpha_i} Y_{i,\beta_i} - X_{i,\beta_i} Y_{i,\alpha_i}.$$

Over $\mathbb{C}(s, t)$, the orthogonal reflection

$$\frac{1}{s^2 + t^2} \begin{pmatrix} s^2 - t^2 & 2st \\ 2st & t^2 - s^2 \end{pmatrix}$$

fixes $s\zeta_1 + t\zeta_2$ and negates each Δ_i . It preserves the joint covariance, so the moment of

$$(s\zeta_1 + t\zeta_2)^{p+1} \prod_{i \in \Omega} \Delta_i$$

is zero: there are an odd number of determinants. Extracting $s^p t$ and dividing by its nonzero binomial factor $p+1$ gives

$$\sum_{I \subseteq \Omega} (-1)^{|I|} H_p(L)(w_I) \Phi(L)(w'_I) = 0, \quad (3)$$

where w_I uses β on I and α elsewhere, and w'_I is complementary. The field $\mathbb{C}(s, t)$ consists of rational functions in two auxiliary parameters. The resulting identity is polynomial, so it also holds when $s^2 + t^2 = 0$, where the displayed reflection has a zero denominator. No operation on the original graph is being made.

A common pure factor and vanishing mixed coefficients. With constant local pair a, b , (3) says

$$f_b[a^\Omega] H_p = f_a[b^\Omega] H_p.$$

Two independent linear functions f are coprime in $\mathbb{C}[D]$: they have no nonconstant common polynomial factor. Hence for $p = 2r + 1$ there is a common homogeneous polynomial q_r of degree $2r$ such that $[h^\Omega] H_p = f_h q_r$ for every h ; $q_0 = 1$. For a mixed word w avoiding an active colour h , choose $\alpha_i = w_i, \beta_i = h$. The only constant complementary word in (3) is h^Ω , giving $f_h[w] H_p = 0$. Polynomial cancellation proves all binary mixed coefficients zero, even at individual rows where $f_h(L) = 0$.

Rotation and finite degree. Define finite polynomials

$$\Psi(L) = \sum_{r=0}^m \frac{H_{2r+1}(L)}{(2r+1)!}, \quad g(L) = \sum_{r=0}^m \frac{q_r(L)}{(2r+1)!}.$$

On any binary palette a, b , the projection is therefore

$$\pi_{ab} \Psi(L) = g(L)(f_a(L)a^\Omega + f_b(L)b^\Omega).$$

Use $K = K_\Omega$ for the ordered colour pair (a, b) , as defined in Section 2.6. The shifted symbols are $X_{i,h} + L_i(h)$ and $Y_{i,h} + M_i(h)$. The shifted-Wick identity shows that $K(\Psi(L), \Psi(M))$ is invariant under simultaneous $\text{SO}(2)$ rotation of the two shift rows: the joint pairing table and each determinant are unchanged. Choose linearly independent f_a, f_b . This pairing is

$$(f_a(L)f_b(M) - f_b(L)f_a(M))g(L)g(M).$$

The determinant factor is a nonzero polynomial and is rotation invariant. Cancel it in $\mathbb{C}[D \times D]$, then rotate by 45 degrees and set $M = 0$. Since g is even and $g(0) = 1$,

$$g(L/\sqrt{2})^2 = g(L).$$

If g had positive degree, the degrees of the two sides would differ. Thus $g = 1$, and all q_r with $r \geq 1$ vanish. Together with (3), this proves whole binary vanishing through the terminal degree $2m + 1$.

For completeness, *polarization* here means substituting $L = t_1 L_1 + \dots + t_p L_p$ and extracting the coefficient of $t_1 \dots t_p$. That coefficient in L^p is $p! L_1 \dots L_p$. Dividing by $p! \neq 0$ proves the stated vanishing for products of arbitrary rows from the same D ; see also [14, Sections 4.4–4.5] for this standard operation. $\square$

For a matching polynomial satisfying (1), the incident rows in Section 3.1 have responses $\tau_h h^{V \setminus \{p\}}$. Their span therefore satisfies (2) with three linearly independent f_h . Thus Lemma 3.1 applies at every original root, before diagonalization.

4 Deleting a second vertex and eliminating bichromatic edges

We next extract the colour of one more vertex from the odd response. The retained vertex set then has even size. The resulting identity says that two incident-row insertions have zero pure coefficient. Applied to deletion matching sums, this will force every aggregate bichromatic edge weight to zero.

4.1 The identity after omitting a vertex

Lemma 4.1 (Even omission). *In the setting of (2), omit a vertex q , put $S = \Omega \setminus \{q\}$, and write*

$$\begin{aligned} R &= Q + \sum_h h_q V_h, & L &= U + \sum_h d_h h_q, \\ E(U) &= \sum_{r=0}^m \frac{U^{2r}}{(2r)!} Q^{[m-r]}, & F &= E(0). \end{aligned}$$

Here Q consists of edges of R lying in S , and V_h is the incident row at q in colour h . The row $U = U(L)$ is L restricted to S , while $d_h = d_h(L) = L_q(h)$ is a scalar. Thus E is the shifted matching polynomial $\mathcal{F}_{S,Q}$ from Section 2.6, and $F = Q^{[m]}$ is its value with no row insertions. For any $L_1, L_2 \in D$ and any active colour k ,

$$[k^S] U(L_1) U(L_2) Q^{[m-1]} = 0.$$

This includes $m = 1$.

Proof. The full coefficient of $q = h$ in $\Psi(L)$ is

$$d_h E(U) + E_{V_h}(U),$$

where the derivative convention is that of Section 2.6, with Q fixed. Fix the target colour k and choose another active colour $h \neq k$. Lemma 3.1 gives

$$\pi_{hk}(d_h E(U) + E_{V_h}(U)) = f_h(L)h^S.$$

We use $K = K_S$ with ordered colours (h, k) . By definition, its arguments are projected by π_{hk} ; no identity of the omitted three-colour coefficients is asserted.

Because $|S|$ is even, K is symmetric and $K(Z, h^S) = [k^S]Z$. For any fixed row direction V on S , the same two-replica rotation gives

$$\begin{aligned} K(E(U), E(U)) &= K(F, E(\sqrt{2}U)), \\ K(E(U), E_V(U)) &= \frac{1}{\sqrt{2}}K(F, E_V(\sqrt{2}U)). \end{aligned} \tag{4}$$

The second identity follows by differentiating the rotation identity at shift rows $U, U + tV$: the second rotated shift is zero at $t = 0$ and $E'(0) = 0$. Substitute

$$\pi_{hk}E_{V_h}(U) = f_h h^S - d_h \pi_{hk}E(U)$$

and the corresponding identity for $\sqrt{2}L$ into the second equation of (4). The first cancels the d_h terms, leaving

$$f_h(L)[k^S](E(U) - F) = 0.$$

Cancel the nonzero polynomial f_h . Separating homogeneous degrees and polarizing yields

$$[k^S]U(L_1)U(L_2)Q^{[m-1]} = 0. \tag{5}$$

□

4.2 From deletion amplitudes to diagonal edge blocks

For a fixed colour h , a *hafnian cofactor* is the matching amplitude obtained by deleting two vertices in that colour. This name does not assert an adjugate or inverse identity. We will prove the needed matrix identity using the omission lemma. All matrices in the following proof have their rows and columns indexed by the original vertex set V .

Lemma 4.2 (Global diagonal reduction). *For every pair of distinct vertices p, r , the colour-indexed matrix $(A_{pr}(i, h))_{i, h \in \{a, b, c\}}$ of a matching polynomial satisfying (1) is diagonal: $A_{pr}(i, h) = 0$ whenever $i \neq h$.*

Proof. For a fixed colour h , define

$$\begin{aligned} M_h[p, r] &= A_{pr}(h, h) & (p \neq r), \quad M_h[p, p] &= 0, \\ C_h[r, q] &= \text{haf}(M_h[V \setminus \{r, q\}]) & (r \neq q), \quad C_h[q, q] &= 0, \\ B_{ih}[p, r] &= A_{pr}(i, h) & (p \neq r), \quad B_{ih}[p, p] &= 0. \end{aligned}$$

Consider the colouring assigning i to p and h to every other vertex. Expanding its matching sum at p gives

$$(B_{ih}C_h)[p, p] = \delta_{i,h}\tau_h.$$

Here $\delta_{i,h}$ is one if $i = h$ and zero otherwise. To compute the off-diagonal entries, take $p \neq q$ and expand each $C_h[r, q]$ at its retained vertex p :

$$\begin{aligned} (B_{ih}C_h)[p, q] &= \sum_{\substack{r,s \in V \setminus \{p,q\} \\ r \neq s}} A_{pr}(i, h) A_{ps}(h, h) \text{haf}(M_h[V \setminus \{p, q, r, s\}]) \\ &= 0. \end{aligned} \tag{6}$$

The last expression is exactly (5) for two rows of the original p family after omitting q . Both sums are ordered; there is no factor $1/2$. Consequently

$$B_{ih}C_h = \delta_{i,h}\tau_h I.$$

Taking $i = h$ proves C_h invertible. Taking $i \neq h$ then proves $B_{ih} = 0$. Every original edge block is therefore diagonal. This uses neither cofactor nonvanishing assumptions nor a change of basis. $\square$

5 The supported-edge identity

The preceding section proves Step 1 of Section 1.4: only monochromatic edges remain. We now prove Step 2. Fix a supported edge pq in one colour. We must show that the matching subtotal containing pq equals the entire amplitude of that colour.

The proof follows the two ways to match the roots: directly to one another, or separately to retained vertices. The original equations constrain a matrix recording these possibilities in two formal copies. Orthogonal symmetry and the row identities restrict that matrix to an identity part and a rank-one part. Two polynomial differential equations then determine both parts, giving the required subtotal.

5.1 Fixed graph data and the two-root expansion

Let B, H be the symmetric zero-diagonal weight matrices for two active colours after diagonal reduction. Write b_i, h_i for the corresponding site variables at vertex i ; these are the two retained colours. Fix roots $p \neq q$, put $S = V \setminus \{p, q\}$ and $N = |S|$, and define

$$Q = \sum_{\substack{i < j \\ i, j \in S}} (B_{ij}b_i b_j + H_{ij}h_i h_j).$$

This is the original edge polynomial restricted to the retained vertices and the two colours. It need not satisfy a GHZ condition. The fixed scalar data are

$$\begin{aligned} d &= B_{pq}, & e &= H_{pq}, & \beta &= \text{haf}(B), \\ \eta &= \text{haf}(H), & \alpha &= \text{haf}(B[S]). \end{aligned}$$

We assume $d \neq 0$, since pq is supported in colour B . The full amplitudes β, η are nonzero; e and the deletion amplitude α may vanish at this stage. The four fixed rows are

row	root	colour	definition on S
x	p	B	$\sum_{i \in S} B_{pi} b_i$
y	q	B	$\sum_{i \in S} B_{qi} b_i$
u	p	H	$\sum_{i \in S} H_{pi} h_i$
v	q	H	$\sum_{i \in S} H_{qi} h_i$

For a shift row L on S , set

$$E(L) = \mathcal{F}_{S,Q}(L) = \sum_{j=0}^{N/2} \frac{L^{2j}}{(2j)!} Q^{[N/2-j]}, \quad F = E(0).$$

Each term pairs some retained vertices using Q and supplies a shift factor at every remaining vertex. All row derivatives use the convention of Section 2.6.

Theorem 5.1 (Supported endpoint identity). *Under the original target equation (1) and the diagonal reduction of Lemma 4.2, every supported edge $B_{pq} \neq 0$ in an active colour satisfies*

$$\beta = B_{pq} \text{haf}(B[V \setminus \{p, q\}]).$$

Lemma 5.2 (Two-root matching identities). *For arbitrary scalar parameters s, t, λ, μ , the fixed rows and polynomial E above satisfy*

$$\begin{aligned} E_y(sx + tu) + dsE(sx + tu) &= s\beta b^S, \\ E_v(sx + tu) + etE(sx + tu) &= t\eta h^S, \\ E_x(\lambda y + \mu v) + d\lambda E(\lambda y + \mu v) &= \lambda\beta b^S, \\ E_u(\lambda y + \mu v) + e\mu E(\lambda y + \mu v) &= \mu\eta h^S. \end{aligned} \tag{7}$$

Proof. The original row at root p , restricted to the two colours, is $s(db_q + x) + t(eh_q + u)$. Apply Lemma 3.1 and extract the colour b at the other root q . A direct edge pq contributes $dsE(sx + tu)$. If q is matched into S , its incident row y is inserted, contributing $E_y(sx + tu)$. The original pure amplitude is $s\beta$. This proves the first identity in (7). Exchanging the colour or the two roots proves the other three identities.

The highest coefficient uses the terminal original odd response of degree $N + 1$. All identities therefore hold in the full polynomial ring of shift parameters, with the graph weights fixed. $\square$

Corollary 5.3 (Specialized row identities). *The same polynomial satisfies*

$$\begin{aligned} E_u(\lambda y) &= 0, & E_v(sx) &= 0, \\ E_{yu}(sx) &= -dsE_u(sx), \\ E_{uv}(sx) + eE(sx) &= \eta h^S, \\ E_{xy}(0) + dF &= \beta b^S. \end{aligned} \tag{8}$$

Proof. Set $t = 0$ in the second identity of (7) and $\mu = 0$ in the fourth to obtain the first line. Differentiate the first two identities with respect to t , then set $t = 0$, to obtain the next two relations. Differentiating the first identity with respect to s at $s = t = 0$ gives the last relation. $\square$

5.2 A matrix of two-copy insertion coefficients

Use $K = K_S$ with ordered colours (b, h) . It is symmetric because N is even, and $K(b^S, h^S) = 1$. The polynomial $K(E(L_1), E(L_2))$ is invariant under the full $O(2)$ action on the two shift rows. The Wick proof is as in Section 3; a determinant -1 transformation contributes $(-1)^N = 1$.

Group eight independent scalar indeterminates into four two-component columns P, Q_c, R_c, S_c . The subscript c labels a column; it is not a graph colour. The entries specify coefficients in copies 1 and 2. For $i \in \{1, 2\}$, define the shift row

$$L_i = P_i x + (Q_c)_i y + (R_c)_i u + (S_c)_i v.$$

The columns can later be specialized to any complex values, including zero; they are not assumed linearly independent. Define the scalar polynomials and matrices

$$\begin{aligned} G &= K(E(L_1), E(L_2)), & f &= G|_{R_c=S_c=0}, \\ M_{ij} &= \partial_{(R_c)_i} \partial_{(S_c)_j} G|_{R_c=S_c=0}, & T &= M + efI. \end{aligned}$$

The entry M_{ij} records an insertion of the H -row u in copy i and of the H -row v in copy j . When both insertions belong to the same copy, they represent the two roots being matched into S . There is one more possibility: the roots are matched directly to each other by the H -edge of weight e , leaving the retained polynomial unchanged. That contribution is ef on each diagonal entry. Thus the correction in T combines the two terms of the original matching expansion, just as $E_{uv} + eE$ does in (8). There is no direct edge between distinct copies.

Here $I = I_2$, and M, T are 2×2 matrices whose indices refer to copies, not graph vertices. We call T the corrected insertion matrix. The scalar G is invariant when an orthogonal matrix O acts on all four columns. Its differentiated matrix therefore satisfies

$$M(OP, OQ_c) = OM(P, Q_c)O^T,$$

and similarly for T . This transformation law is called matrix *covariance*; it differs from the pairing coefficients called covariances earlier. By (8),

$$M_{12} = K(E_u(P_1x + (Q_c)_1y), E_v(P_2x + (Q_c)_2y))$$

is divisible by $P_1(Q_c)_2$; likewise M_{21} is divisible by $P_2(Q_c)_1$.

5.3 The matrix form forced by symmetry and divisibility

Use the bilinear dot product $P \cdot Q_c = P_1(Q_c)_1 + P_2(Q_c)_2$, with no conjugation. The three *Gram coordinates* are the pairings of the two columns with themselves and each other. The next lemma says that the insertion matrix has only two scalar polynomial coefficients to determine. The matrix PQ_c^T is an outer product and has rank at most one.

Lemma 5.4 (Polynomial form of the corrected insertion matrix). *For the corrected insertion matrix just defined,*

$$\begin{aligned} T &= \varphi I + \psi PQ_c^T, \\ \varphi, \psi &\in \mathbb{C}[\sigma, \tau, \rho], \\ \sigma &= P \cdot P, \quad \tau = Q_c \cdot Q_c, \quad \rho = P \cdot Q_c. \end{aligned} \tag{9}$$

Proof sketch; complete proof in Appendix A. The off-diagonal divisibility holds in every orthogonal coordinate system. It implies that MP is parallel to P and $M^T Q_c$ is parallel to Q_c . On the set where the needed bilinear products are nonzero, these conditions give a scalar-identity part and one rank-one part. The same divisibility makes their coefficients polynomials, so the identity extends to all parameter values. Those scalar coefficients are invariant under simultaneous orthogonal transformations. The classical scalar invariant theorem then expresses them in the three Gram coordinates [14, Section 10.2(a)]. Appendix A supplies both the matrix argument and a direct proof of this scalar specialization. $\square$

Orthogonal covariance alone would allow other matrices, for example PP^T . The divisibility obtained from the original matching equations is what restricts the matrix to (9). The classical invariant theorem is used only for its scalar coefficients.

5.4 Finite polynomial degree determines the matrix

Lemma 5.5 (Polynomial differential equations). *Let $\mathcal{R}$ be a commutative integral domain, let $d \in \mathcal{R}$ be nonzero, and let $p \in \mathcal{R}[z]$. If $p' + dp = 0$, then $p = 0$. If $p' + dp = k$ for $k \in \mathcal{R}$, then p is constant and $dp(0) = k$.*

Proof. If $p \neq 0$ has degree r , the coefficient of z^r in $p' + dp$ is d times the leading coefficient of p , which is nonzero. This excludes a homogeneous equation. In the inhomogeneous equation it excludes $r > 0$; substituting a constant polynomial gives $dp(0) = k$. $\square$

The coefficient domain may itself be a polynomial ring. This elementary step, including its endpoint cancellation consequence, is formalized in *PolynomialODE.lean* [15].

Proof of Theorem 5.1. We first eliminate the rank-one coefficient ψ , then determine the identity coefficient φ . A subscript ρ denotes formal partial differentiation in the Gram coordinate ρ , with σ, τ held fixed. For either Gram-polynomial coefficient θ , the chain rule gives

$$\partial_{(Q_c)_1} \theta = 2(Q_c)_1 \theta_\tau + P_1 \theta_\rho.$$

Thus on the slice $(Q_c)_1 = 0$, this derivative is $P_1 \theta_\rho$. This is how differentiation in a copy parameter yields the following equations in the Gram coordinate ρ .

The rank-one coefficient vanishes. At $(Q_c)_1 = 0$, differentiate M_{12} in $(Q_c)_1$ and use (8). One obtains

$$\partial_{(Q_c)_1} M_{12} \Big|_{(Q_c)_1=0} = -dP_1 M_{12} \Big|_{(Q_c)_1=0}, \quad P_1^2 (Q_c)_2 (\psi_\rho + d\psi) = 0.$$

The substitution

$$(\sigma, \tau, \rho) = (P_1^2 + P_2^2, (Q_c)_2^2, P_2(Q_c)_2)$$

is injective on polynomial rings: every $\tau \neq 0$ and arbitrary σ, ρ is attained over $\mathbb{C}$. Thus $\psi_\rho + d\psi = 0$. Apply Lemma 5.5 over $\mathbb{C}[\sigma, \tau]$, with polynomial variable ρ . Since $d \neq 0$, it gives $\psi = 0$, and hence $T = \varphi I$.

The identity coefficient is constant. Write

$$Z = E_{uv}(P_2 x + (Q_c)_2 y) + eE(P_2 x + (Q_c)_2 y).$$

The full expression is $T_{22} = K(E(P_1x + (Q_c)_1y), Z)$. Differentiate this expression in $(Q_c)_1$, then set $(Q_c)_1 = 0$. Equation (7) gives

$$\partial_{(Q_c)_1} T_{22} \Big|_{(Q_c)_1=0} = P_1 \beta K(b^S, Z) - dP_1 T_{22} \Big|_{(Q_c)_1=0}.$$

The pure- H coefficient of Z is η : positive-degree B -colour row insertions cannot contribute to an all- H word, so it is the same as at zero, where (8) gives $E_{uv}(0) + eF = \eta h^S$. Thus $K(b^S, Z) = \eta$. Using (9) and the same injective substitution proves

$$\varphi_\rho + d\varphi = \beta\eta.$$

Lemma 5.5, again over $\mathbb{C}[\sigma, \tau]$, gives $\varphi = \beta\eta/d$, independent of all three Gram variables. At the origin,

$$\varphi(0) = K(F, E_{uv}(0) + eF) = \eta K(F, h^S) = \eta\alpha.$$

Since $\eta \neq 0$, this proves

$$\beta = B_{pq} \text{haf}(B[V \setminus \{p, q\}]) \quad \text{whenever } B_{pq} \neq 0. \quad (10)$$

Throughout the argument the edge weights are fixed. All differentiations are in the auxiliary columns P, Q_c , and the polynomial identities are finite. $\square$

6 Every colour is a perfect matching

Corollary 6.1 (Matching rigidity of every active colour). *Each active colour graph of a matching polynomial satisfying (1) is a weighted perfect matching.*

Proof. The ordinary hafnian expansion at a vertex p says

$$\beta = \sum_{q \neq p} B_{pq} \text{haf}(B[V \setminus \{p, q\}]).$$

By (10), each supported edge contributes β , and unsupported edges contribute zero. Since $\beta \neq 0$, the number of supported B neighbors of p is exactly one. This holds at every vertex, and for each of the three colours by choosing any other active colour as H . $\square$

Corollary 6.2 (Disjoint colour matchings and unique contributions). *No unordered vertex pair belongs to two colour matchings. Every vertex colouring receives a contribution from at most one perfect matching.*

Proof. Every vertex has exactly one incident aggregate edge of each active colour. A prescribed vertex colouring therefore supports at most one matching. If a matching exists, its contribution is the product of its nonzero edge weights and cannot cancel.

Two colour matchings cannot share a vertex pair pq : choosing that edge in colour B and the other matching in colour H off p, q gives a mixed colouring with a unique, nonzero contribution. This contradicts (1). Their union is consequently a simple cubic graph with one edge of each colour at every vertex. Any mixed perfect matching in this union would likewise violate (1). $\square$

7 The remaining graph lemma

The following lemma is a special case of Bogdanov's obstruction [6], also recorded in [5, Theorem 7]. At this point, each colour is already a perfect matching, and no vertex colouring can receive cancelling matching contributions. We give an adapted version of Bogdanov's alternating-cycle and minimal-arc proof for this special case.

Here a perfect matching is *mixed* if it uses edges from at least two of the colour classes F, G, H . Since every edge is monochromatic, this is equivalent to inducing a mixed vertex colouring.

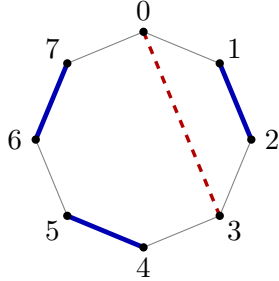
Lemma 7.1 (Three-matching obstruction). *Three pairwise edge-disjoint perfect matchings F, G, H on $n > 4$ vertices have a mixed perfect matching in their union.*

Proof. Suppose otherwise. The union $F \cup G$ must be a single alternating Hamilton cycle C , meaning a cycle visiting every vertex once: otherwise choosing F on one component and G on the others already gives a mixed matching. Label its vertices $0, \dots, n-1$ cyclically. The matching H consists of chords, namely edges joining two vertices of C that are not consecutive on it.

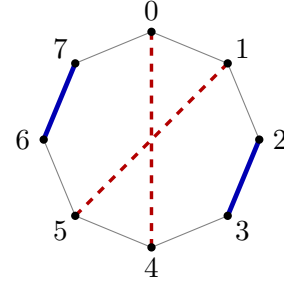
Figure 3 illustrates the next two constructions. A chord with endpoints of opposite parity leaves two even paths when its endpoints are deleted. Matching those paths along C and adding the chord gives a mixed perfect matching. Hence all H chords join equal parities. The two parity classes must each have even size; otherwise H is impossible.

An even-even chord and an odd-odd chord whose endpoints interlace (occur alternately around the cycle) leave four even paths on deleting their endpoints. Those paths and the two chords give a mixed perfect matching when $n > 4$. Thus no such pair interlaces.

Choose a chord and one of its sides with the fewest vertices strictly inside, over all chords and both sides. Its endpoints have equal parity, so its interior contains a vertex of the opposite parity. Every vertex of that opposite parity in the interior must be H -matched to another interior vertex; a partner outside would give an interlacing opposite-parity chord. Therefore some H chord lies strictly inside the chosen arc. One side of that chord has fewer interior vertices, a contradiction. $\square$



(a) One opposite-parity chord:
 $H : \{03\}$, cycle edges $\{12, 45, 67\}$.



(b) Interlacing equal-parity chords:
 $H : \{04, 15\}$, cycle edges $\{23, 67\}$.

Figure 3: Two ways to construct a mixed perfect matching, illustrated on an eight-cycle. Dashed edges belong to H ; heavy solid edges are chosen from the cycle $F \cup G$. Together they cover every vertex exactly once. Only the selected H -edges are drawn. The crossing in (b) is not a vertex. These are local configurations used in the proof, not GHZ constructions.

This supplies a mixed perfect matching, contradicting Section 6. Equation (1) is impossible for every even $n > 4$, completing the proof of Theorem 2.1. The graph interpretation in Section 2 and restriction to three active colours now give Theorem 1.2.

8 Boundary cases and formal verification

For $n = 4$, the three matchings of K_4 give the permitted three-colour construction. In Section 7, a pair of interlacing chords then exhausts the vertices and gives the pure third matching, so no contradiction is asserted. Sections 3–6 remain valid at $n = 4$, including their terminal response coefficients. For $n = 2$, Lemma 3.1 does not apply and arbitrary target dimension is possible. Odd n has no perfect matchings in this model.

The normalized equation-system statement is formalized in Lean for every even $N \geq 6$ and every $D \geq 3$, with arbitrary complex aggregate endpoint-colour weights [15]. The real-weight transfer is also checked. Lemma 2.2 and the aggregation in Section 1.4 explain the mathematical bridge to the graph statement used here. Appendix B gives the exact theorem, verification boundary, reproduction commands, and written-proof provenance.

A Polynomial form of the insertion matrix

This appendix supplies the coordinate details deferred from Lemma 5.4. Every pairing and orthogonal frame below uses the complex bilinear dot product defined in Section 5.

Proof of Lemma 5.4. We first work where all five quantities

$$\sigma, \quad \tau, \quad \rho, \quad \det[P \ Q_c], \quad P_1(Q_c)_2$$

are nonzero. Here $[P \ Q_c]$ is the matrix with columns P, Q_c . This restriction will be removed by polynomial divisibility below. In an orthonormal frame where $P_1 = 0$, $M_{12} = 0$, so MP is parallel to P . In a frame where $(Q_c)_2 = 0$, $M_{12} = 0$, so $M^T Q_c$ is parallel to Q_c . Such frames exist because the relevant bilinear squares σ, τ are nonzero. Their eigenvalues agree because $\rho \neq 0$. For two independent columns in dimension two, these conditions force

$$M = \varphi_0 I + \psi P Q_c^T.$$

Indeed, M minus the common eigenvalue times I annihilates P and has left kernel Q_c ; its rank-one form is a multiple of $(P \cdot Q_c)I - P Q_c^T$. The off-diagonal divisibility proved in the main text shows that

$$\psi = \frac{M_{12}}{P_1(Q_c)_2}, \quad \varphi_0 = M_{11} - \psi P_1(Q_c)_1$$

are polynomials. The resulting identity, proved whenever the excluded polynomials are nonzero, therefore holds identically, including at their zeros. Uniqueness and covariance make φ_0, ψ scalar $O(2)$ invariants, and the same is true of $\varphi = \varphi_0 + ef$.

The scalar invariant-ring assertion is the two-vector case of the first fundamental theorem for the orthogonal group; see [14, Section 10.2(a)]. For completeness, we prove this particular case directly. In coordinates $p_+ = P_1 + iP_2$, $p_- = P_1 - iP_2$, and $q_{\pm} = (Q_c)_1 \pm i(Q_c)_2$, where $i^2 = -1$, a matrix in $SO(2)$ multiplies the plus coordinates by a nonzero scalar z and the minus coordinates by z^{-1} . Its invariant monomials are generated by $\sigma, \tau, r = p_+ q_-$, $s = p_- q_+$, with $rs = \sigma\tau$. Reflection exchanges r, s . Symmetric polynomials reduce to $r + s = 2\rho$ and $rs = \sigma\tau$, giving $\mathbb{C}[\sigma, \tau, \rho]$. There is no polynomial relation among the three Gram coordinates: when $\tau \neq 0$, choose $Q_c = (0, \sqrt{\tau})^T$, $P_2 = \rho/\sqrt{\tau}$, and $P_1^2 = \sigma - \rho^2/\tau$. This realizes arbitrary σ, ρ ; a polynomial relation would therefore vanish identically. $\square$

B Formal artifact and proof provenance

The historical sources collected in [16] are internal research notes from this repository. Their proofs have been included here, so these citations identify provenance rather than replace mathematical steps. The published external inputs are the graph formulation and combinatorial obstruction [3, 5]; the pairing and invariant-theory references [13, 14] provide background for identities also proved in the text.

The Lean development in [15] proves nonexistence for every even $N \geq 6$ and every $D \geq 3$, with arbitrary complex aggregate endpoint-colour weights and the normalized equations

$$W(\kappa) = \begin{cases} 1, & \kappa \text{ is constant,} \\ 0, & \kappa \text{ is mixed.} \end{cases}$$

The final declaration is `UpstreamFullProof.eqSystem_no_solution_ge6_ge3` in `FullProof.lean`, within the namespace `KrennAllOrders`. An adapter identifies the local matching sums with the exact `MonochromaticQuantumGraph.WeightsN` and `EqSystemN` definitions in the pinned Formal Conjectures source. The final theorem has no remaining source-reduction or structural hypothesis.

The checked proof covers the complete chain: the matching model, copy symmetries, response and omission identities, diagonal reduction, the polynomial matrix argument, the supported-edge identity, and the graph obstruction. The real-to-complex transfer and the corresponding real-weight corollaries are also formalized.

The dependency audit reports only three standard axioms: `propext`, `Classical.choice`, and `Quot.sound`. There are no proof holes or additional axioms in the final dependencies.

The final Lean statement uses aggregate weights and unit pure amplitudes. Sections 1.4 and 2.3 give the explicit mathematical translations from the multigraph formulation and arbitrary nonzero pure amplitudes used here. These explanatory translations should be distinguished from the exact interface of the linked Lean theorem. The artifact includes the pinned toolchain, source hashes, axiom reports, and commands for reproducing both the local proof and the upstream adapter.

For the local proof, install Lean through `elan` and run from the repository root:

```
cd formal/all-orders
lake exe cache get
python3 verify.py
```

The toolchain is Lean 4.33.1; the package manifest fixes its `mathlib` dependency. To check the exact upstream interface, prepare the Formal Conjectures checkout at the revision recorded in `formal/upstream-adapter/README.md`, then run from the repository root:

```
cd formal/upstream-adapter
python3 verify.py --upstream /path/to/formal-conjectures
```

This second command checks the pinned upstream revision and compiles the adapter, final complex theorem, and real corollaries. The linked reproduction instructions in [15] specify the checkout and cache commands in full.

The written argument is based on the frozen mathematical source in [16]. The associated agent audits are internal research records; they do not constitute external peer review. This manuscript develops its exposition independently of those historical records.

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